

filters and stability

1 Filters

- signals as sine functions
- linear, time invariant, causal filters
- convolutions
- removing a signal

2 Stability

- bounded inputs and bounded outputs
- amplitude gain and phase shift

MCS 472 Lecture 7
Industrial Math & Computation
Jan Verschelde, 28 January 2026

filters and stability

1 Filters

- signals as sine functions
- linear, time invariant, causal filters
- convolutions
- removing a signal

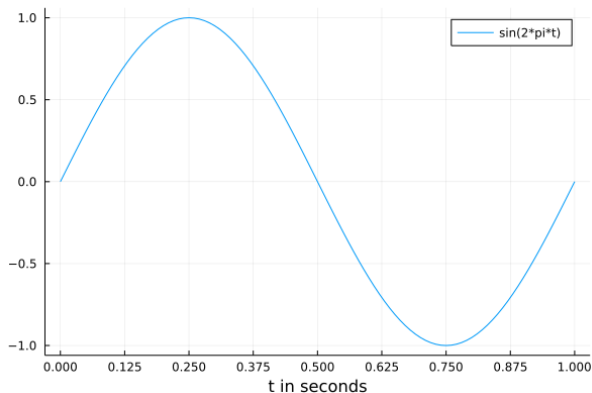
2 Stability

- bounded inputs and bounded outputs
- amplitude gain and phase shift

signals as sine functions

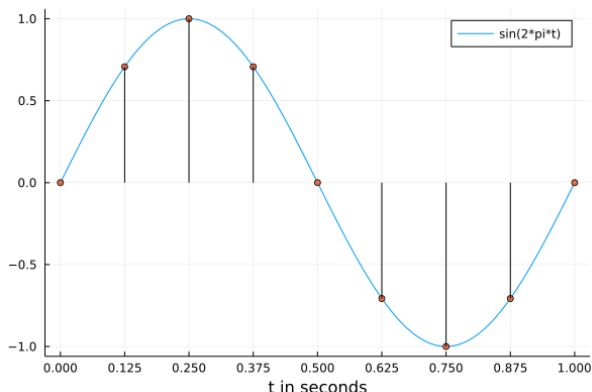
Every signal can be written as a linear combinations of sine functions running at increasing frequencies, as $\sin(2\pi kt)$, $k = 1, 2, \dots$

Frequencies are measured in Hertz (Hz): n Hz = n cycles per second.



sampling at 8 Hertz

Sampling at 8 Hertz means taking 8 equidistant samples per second.



Our signal is $\{u_0, u_1, u_2, u_3, u_4, u_5, u_6, u_7, u_0, \dots\}$,
where $u_k = \sin(2\pi k/8)$, $k = 0, 1, 2, \dots, 7$.

the z-transform of a sine function

Exercise 1:

What is the z-transform of a one Hertz sine function sampled at eight Hertz?

filters and stability

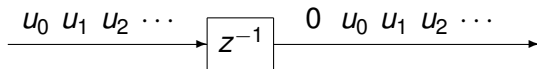
1 Filters

- signals as sine functions
- linear, time invariant, causal filters
- convolutions
- removing a signal

2 Stability

- bounded inputs and bounded outputs
- amplitude gain and phase shift

unit delay as a filter



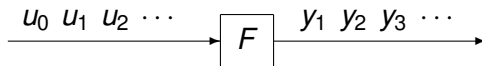
The input $u = \{u_0, u_1, u_2, \dots\}$ has z-transform $U(z) = \sum_{k=0}^{\infty} u_k z^{-k}$.

The output $y = \{0, u_0, u_1, u_2, \dots\}$ has z-transform $Y(z)$.

$$\begin{aligned} Y(z) &= \sum_{k=1}^{\infty} u_{k-1} z^{-k} \\ &= z^{-1} \sum_{k=1}^{\infty} u_{k-1} z^{-(k-1)} = z^{-1} \sum_{k=0}^{\infty} u_k z^{-k} \\ &= z^{-1} U(z) \end{aligned}$$

We can view the delay of a signal as a *filter*.

the definition of a filter



Definition (linear, time invariant, causal filter)

A **filter** F turns input $u = \{u_k\}_{k=0}^{\infty}$ into output $y = \{y_k\}_{k=0}^{\infty}$ and is

- 1 **linear:** $F(\alpha u^{(1)} + \beta u^{(2)}) = \alpha F(u^{(1)}) + \beta F(u^{(2)})$,
- 2 **time invariant:** $F\left(\{u_{k+k_0}\}_{k=0}^{\infty}\right) = \{y_{k+k_0}\}_{k=0}^{\infty}$, $k > k_0$, and
- 3 **causal:** $u_k = 0, k < k_0 \Rightarrow y_k = 0, k < k_0$.

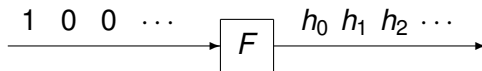
the unit delay as a filter

Exercise 2:

Show that the unit delay $F = z^{-1}$ is linear, time invariant, and causal.

the unit impulse

The *unit impulse* is $\delta = \{1, 0, 0, \dots\}$.



The *impulse response* is $h = F\delta = \{h_0, h_1, h_2, \dots\}$.

Theorem

A linear, time invariant, causal filter is entirely determined by the impulse response.

For input u , the k -th element in the output y is

$$y_k = u_k h_0 + u_{k-1} h_1 + \dots + u_1 h_{k-1} + u_0 h_k = \sum_{j=0}^k h_j u_{k-j}.$$

The convolution is denoted by the operator $\star$, as $y = h \star u$.

the transfer function

In the frequency domain, the z-transforms of the input and output are

$$U(z) = \sum_{k=0}^{\infty} u_k z^{-k} \quad \text{and} \quad Y(z) = \sum_{k=0}^{\infty} y_k z^{-k}.$$

The impulse response has as z-transform $H(z) = \sum_{k=0}^{\infty} h_k z^{-k}$.

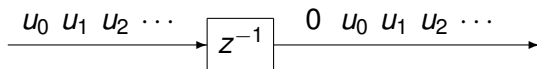
The z-transform of $y = h \star u$ is then

$$Y(z) = H(z)U(z).$$

Definition (the transfer function)

The *transfer function of a filter* is the z-transform of the impulse response.

the transfer function of the unit delay filter



The input $u = \{u_0, u_1, u_2, \dots\}$ has z-transform $U(z) = \sum_{k=0}^{\infty} u_k z^{-k}$.

The output $y = \{0, u_0, u_1, u_2, \dots\}$ has z-transform $Y(z) = z^{-1} U(z)$.

What is the transfer function of this filter?

The transfer function is the z-transform of the impulse response:

- The response of $\{1, 0, 0, \dots\}$ is $\{0, 1, 0, 0, \dots\}$.
- The z-transform of $\{0, 1, 0, 0, \dots\}$ is z^{-1} .

Thus z^{-1} is the transfer function of the unit delay filter.

filters and stability

1 Filters

- signals as sine functions
- linear, time invariant, causal filters
- **convolutions**
- removing a signal

2 Stability

- bounded inputs and bounded outputs
- amplitude gain and phase shift

computing the output of a filter

Denote

$$\begin{aligned}\delta^{(0)} &= \{1, 0, 0, \dots\}, \\ \delta^{(1)} &= \{0, 1, 0, \dots\}, \\ \delta^{(2)} &= \{0, 0, 1, \dots\}, \quad \text{etc.}\end{aligned}$$

then we can write any input u as $u = \sum_{k=0}^{\infty} u_k \delta^{(k)}$.

The output $y = Fu$ can then be written as

$$y = F \left(\sum_{k=0}^{\infty} u_k \delta^{(k)} \right) = \sum_{k=0}^{\infty} u_k F \delta^{(k)},$$

because F is linear.

convolutions

After decomposing any input u as linear combinations of properly delayed unit impulses, we construct a formula for the output.

$$\begin{aligned}y &= u_0 F\delta^{(0)} + u_1 F\delta^{(1)} + u_2 F\delta^{(2)} + \dots \\&= u_0 \{h_0, h_1, h_2, \dots\} + u_1 \{0, h_0, h_1, h_2, \dots\} \\&\quad + u_2 \{0, 0, h_0, h_1, h_2, \dots\} \\&= \{u_0 h_0, u_0 h_1 + u_1 h_0, u_0 h_2 + u_1 h_1 + u_2 h_0, \dots\} \\&= \sum_{k=0}^{\infty} \underbrace{\sum_{j=0}^k u_j h_{k-j}}_{= y_k}\end{aligned}$$

The formula for y_k is a convolution of the input with the impulse response. We proved that the impulse response determines a filter.

filters and stability

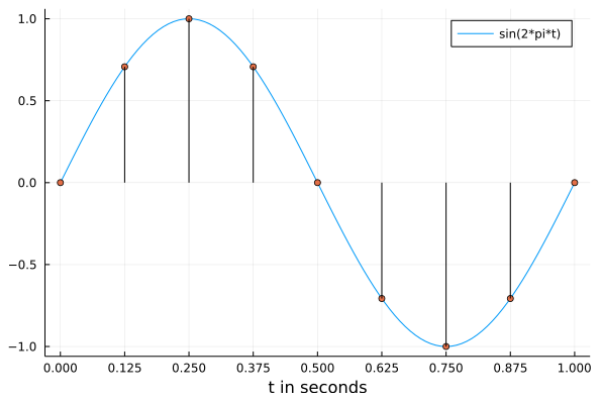
1 Filters

- signals as sine functions
- linear, time invariant, causal filters
- convolutions
- removing a signal

2 Stability

- bounded inputs and bounded outputs
- amplitude gain and phase shift

removing a signal



Our signal is $\{u_0, u_1, u_2, u_3, u_4, u_5, u_6, u_7, u_0, \dots\}$,
where $u_k = \sin(2\pi k/8)$, $k = 0, 1, 2, \dots, 7$.

Observe: $u_5 = -u_1$, $u_6 = -u_2$, $u_7 = -u_3$.

adding a delay to remove the signal

Our signal is $\{u_0, u_1, u_2, u_3, u_4, u_5, u_6, u_7, u_0, \dots\}$,

where $u_k = \sin(2\pi k/8)$, $k = 0, 1, 2, \dots, 7$.

Observe: $u_5 = -u_1$, $u_6 = -u_2$, $u_7 = -u_3$.

Then let $y_0 = u_0 + u_4$, $y_1 = u_1 + u_5$, $y_2 = u_2 + u_6$, $y_3 = u_3 + u_7$.

This results in $y_0 = 0$, $y_1 = 0$, $y_2 = 0$, $y_3 = 0$.

The z-transform of y is $Y(z)$:

$$Y(z) = U(z) + z^{-4}U(z) = (1 + z^{-4}) U(z).$$

The transfer function of the filter is $H(z) = 1 + z^{-4}$.

removing high frequency noise

Noise often happens at high frequency.

Exercise 3:

Give the transfer function of a filter to remove a signal at 60 Hz, sampled at 720 Hz.

- 1 Justify the design of your filter.
- 2 Which signals are also removed by your filter?

filters and stability

1 Filters

- signals as sine functions
- linear, time invariant, causal filters
- convolutions
- removing a signal

2 Stability

- bounded inputs and bounded outputs
- amplitude gain and phase shift

stable filters

Definition

A filter is *stable* if bounded inputs produce bounded outputs.

What are bounded inputs and outputs?

Some examples:

- $x = \{1, 1, 1, 1, 1, 1, 1, \dots\}$ is bounded, and so is any constant signal.
- $x = \{1, 2, 4, 8, 16, 32, 64, \dots\}$ is not bounded.
- $x = \left\{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \dots\right\}$ is bounded.

the transfer function of a stable filter

The impulse response determines a linear, time invariant, causal filter.

Proposition

Let $H(z) = \sum_{k=0}^{\infty} h_k z^{-k}$ be the transfer function of a filter F .

$$F \text{ is stable} \iff \sum_{k=0}^{\infty} |h_k| < \infty.$$

As a corollary, the transfer function $H(z)$ of a stable filter converges absolutely and uniformly for all $|z| \geq 1$ to a continuous function.

filters and stability

1 Filters

- signals as sine functions
- linear, time invariant, causal filters
- convolutions
- removing a signal

2 Stability

- bounded inputs and bounded outputs
- amplitude gain and phase shift

amplitude gain and phase shift

Theorem

Let $H(z) = \sum_{k=0}^{\infty} h_k z^{-k}$ be the transfer function of a filter F .

For input $u = \left\{ u_k = \sin(\omega kT) \right\}_{k=0}^{\infty}$, $y = \left\{ y_k = r \sin(\omega kT + \phi) \right\}_{k=0}^{\infty}$ is the output, where

- $r = |H(e^{i\omega T})|$ is the amplitude gain, and
- $\phi = \arg H(e^{i\omega T})$ is the phase shift.

Interpret the input u as

- $u = \sin(\omega t)$, $\omega = 2\pi n$, at n Hertz,
- sampled at $t = kT$, T is the sampling rate.

Complex numbers appear, the imaginary unit is $i = \sqrt{-1}$.

proof using complex numbers

We look at the response of the signal $s = e^{j\omega T}$.

The input $u = \{s^k\}_{k=0}^{\infty}$. Let us compute $y = h \star u$.

$$y_k = \sum_{j=0}^k h_j u_{k-j} = \sum_{j=0}^k h_j s^{k-j} = s^k \sum_{j=0}^k h_j s^{-j}$$

By the definition of H : $H(s) = \sum_{j=0}^{\infty} h_j s^{-j}$. So, we have:

$$y_k = s^k H(s) - s^k \sum_{j=k+1}^{\infty} h_j s^{-j}.$$

Consider $k \rightarrow \infty$.

complex exponentiation and the sine function

Our input signal is $s = e^{i\omega T}$.

$$\frac{s - s^{-1}}{s - s^{-1}} = \frac{e^{i\omega T} - e^{-i\omega T}}{e^{i\omega T} - e^{-i\omega T}} = \frac{\cos(\omega T) + i \sin(\omega T) - (\cos(\omega T) - i \sin(\omega T))}{2i \sin(\omega T)}$$

It then follows:

$$\sin(\omega T) = \frac{1}{2i} (s - s^{-1})$$

and

$$\sin(k\omega T) = \frac{1}{2i} (s^k - s^{-k}).$$

computing the amplitude gain and phase shift

We derived $y_k = s^k H(s)$. Suppose $H(e^{i\omega T}) = re^{i\phi}$.

What are r and ϕ ?

$$\begin{aligned}\frac{s^k H(s) - s^{-k} H(s^{-1})}{2i} &= \frac{1}{2i} \left(s^k re^{i\phi} - s^{-k} re^{-i\phi} \right), \quad s = e^{i\omega T} \\ &= \frac{1}{2i} \left(e^{ki\omega T + \phi} - e^{-(ki\omega T + \phi)} \right) \\ &= r \sin(k\omega T + \phi)\end{aligned}$$

Then indeed, $r = |H(e^{i\omega T})|$ and $\phi = \arg H(e^{i\omega T})$.

Q.E.D.